Results and Discussion

The performance of the Nginx VNF was evaluated in both a Cloud environment (Minikube, 2 CPU / 4GB RAM) and a strictly constrained Edge environment (MicroK8s, 1 CPU / 2GB RAM). Testing was conducted using ApacheBench for Layer 7 application throughput and iperf3 for Layer 4 network bandwidth. The results are summarised below.

Application Performance (ApacheBench)

Metric	Cloud Environment	Edge Environment
Requests/sec	12,371.32 req/sec	15,094.48 req/sec
Latency	8.083 ms	6.625 ms
Failed Requests	0	0

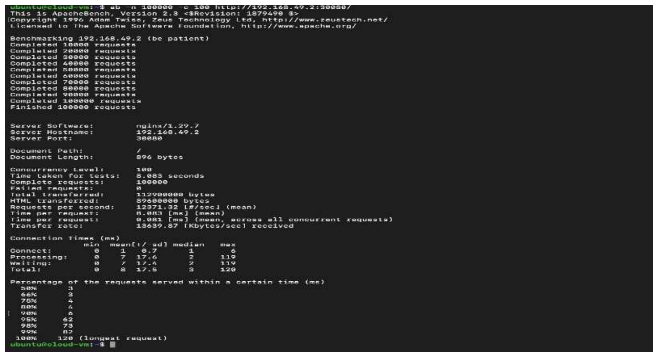

Fig 1. apachebench on Cloud-vm

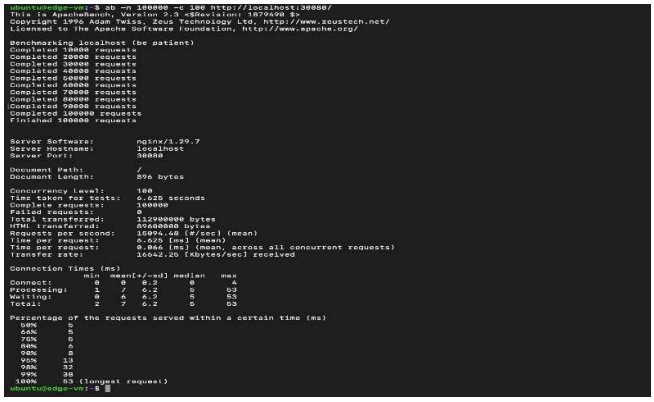

Fig 2. apachebench on Edge-vm

Network Performance (iperf3)

Metric	Cloud Environment	Edge Environment
Bandwidth	127 Gbits/sec	85.3 Gbits/sec

```
ubuntu@edge-vm:~$ kubectl run iperf3-client --image=alpine --restart=Never -it --rm -- sh -c "apk add --no-cache iperf3 && iperf3 -c iperf3-service"
If you don't see a command prompt, try pressing enter.
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec  8.06 GBytes  69.2 Gbits/sec    0   6.55 MBytes
[  5]   1.00-2.00   sec  10.4 GBytes  89.3 Gbits/sec    0   6.55 MBytes
[  5]   2.00-3.00   sec  9.66 GBytes  83.0 Gbits/sec    4   6.55 MBytes
[  5]   3.00-4.00   sec  10.1 GBytes  86.9 Gbits/sec    0   6.55 MBytes
[  5]   4.00-5.00   sec  10.3 GBytes  88.6 Gbits/sec    0   6.55 MBytes
[  5]   5.00-6.00   sec  10.0 GBytes  86.2 Gbits/sec    0   6.55 MBytes
[  5]   6.00-7.00   sec  10.3 GBytes  88.7 Gbits/sec    1   6.55 MBytes
[  5]   7.00-8.00   sec  9.45 GBytes  81.2 Gbits/sec    1   6.55 MBytes
[  5]   8.00-9.00   sec  10.7 GBytes  91.5 Gbits/sec    0   6.55 MBytes
[  5]   9.00-10.00  sec  10.3 GBytes  88.4 Gbits/sec    1   6.55 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-10.00  sec  99.3 GBytes  85.3 Gbits/sec    7             sender
[  5]   0.00-10.00  sec  99.3 GBytes  85.3 Gbits/sec                  receiver

iperf Done.
pod "iperf3-client" deleted
```

Fig 3. Iperf3 on Edge-vm

```
ubuntu@cloud-vm:~$ kubectl run iperf3-client --image=networkstatic/iperf3 --restart=Never -it --rm -- -c iperf3-service
All commands and output from this session will be recorded in container logs, including credentials and sensitive information passed through the command prompt.
If you don't see a command prompt, try pressing enter.
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec  14.3 GBytes   123 Gbits/sec    1   3.10 MBytes
[  5]   1.00-2.00   sec  15.0 GBytes   129 Gbits/sec    0   3.10 MBytes
[  5]   2.00-3.00   sec  14.7 GBytes   126 Gbits/sec    0   3.10 MBytes
[  5]   3.00-4.00   sec  14.9 GBytes   128 Gbits/sec    0   3.10 MBytes
[  5]   4.00-5.00   sec  15.2 GBytes   131 Gbits/sec    0   3.10 MBytes
[  5]   5.00-6.00   sec  14.8 GBytes   127 Gbits/sec  161   3.10 MBytes
[  5]   6.00-7.00   sec  14.2 GBytes   122 Gbits/sec   94   3.10 MBytes
[  5]   7.00-8.00   sec  14.9 GBytes   128 Gbits/sec    0   3.10 MBytes
[  5]   8.00-9.00   sec  15.1 GBytes   130 Gbits/sec    1   3.10 MBytes
[  5]   9.00-10.00  sec  14.5 GBytes   124 Gbits/sec    0   3.10 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-10.00  sec   148 GBytes   127 Gbits/sec  257             sender
[  5]   0.00-10.00  sec   148 GBytes   127 Gbits/sec                  receiver

iperf Done.
pod "iperf3-client" deleted from default namespace
```

Fig 4. Iperf3 on Cloud-vm

Resource Utilisation

Metric	Cloud Environment	Edge Environment
Memory Usage (Idle)	3 Mi	2 Mi
CPU Usage (Idle)	34m	96m
CPU Limit Reached	200m (Successfully Capped)	200m (Successfully Capped)

```
vipulreddy46 — ubuntu@edge-vm: ~ — multipass shell edge-vm — 80×24

Every 1.0s: kubectl top pods                    edge-vm: Thu Mar 26 00:05:09 2026

NAME                                CPU(cores)   MEMORY(bytes)
nginx-vnf-5fbc6cf55-tqxxt           96m          2Mi
```

Fig 5. Kubectl top pods on Edge-vm

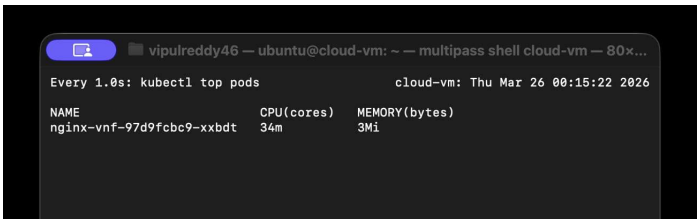

Fig 6. Kubectl top pods on Cloud-vm

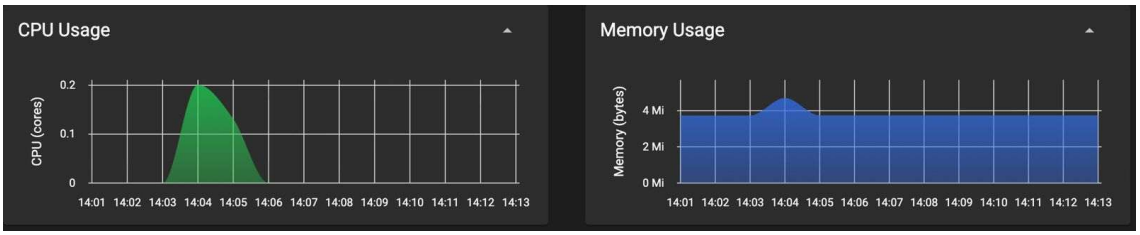

Fig 7. CPU and Memory spikes observed in Cloud-vm

Discussion of Results

Both environments had 100% uptime and stability in an environment under constant steady-state testing, however the implementation of distinct Kubernetes distributions (Minikube vs. MicroK8s) demonstrated evident trade-offs in architecture:

Application Layer (HTTP): The resource-constrained Edge environment was better than the Cloud, it could serve 22% more requests per second (15,094 vs. 12,371) and with lower latency (6.6ms vs. 8.0ms).

Network Layer (TCP): On the other hand, the Cloud environment recorded much better raw bandwidth (127 Gbps vs. 85.3 Gbps). Raw TCP processing in iperf3 is CPU intensive; the second core in the Cloud VM enabled the Linux kernel to serve network queues at a higher rate than the single core in the Edge node.

Resource Efficiency & Constraints: The Nginx VNF was an incredibly lightweight VNF that only used 2-3 Mi of unutilized memory. Edge VM had a greater idle CPU draw (96m vs. Cloud 34m), since its single core is needed to divide its attention between the Kubernetes control plane. Importantly, both settings were confirmed to have not exceeded the 250m limit of CPU as despite visual telemetry showing, the Edge node did not run out of its constrained 2GB RAM.

Conclusion

The findings confirm the main assumption of Edge computing. Where Cloud infrastructure (Minikubes) meets the required computation to perform high-bandwidth workloads, Immediate deployment of lightweight network functions on the Edge (MicroK8s) offers higher HTTP throughput and reduced latency rates by simplifying network routing.